

CRYTALLOGRAPHY

Q1. Derive Expression for Interplanar spacing.

The distance between neighboring Miller planes is the same as the nearest perpendicular distance between the origin and the closest Miller plane. A miller plane ABC shown in Fig.1.32 belongs to the family of planes whose miller indices are (hkl). ON is normal to plane drawn from origin and is equal to d_{hkl} . The d_{hkl} represents interplanar spacing of family of plane (hkl). Let α , β , and γ be the angles between coordinate axes X, Y, Z and normal ON respectively. These angles are different from interfacial angles.

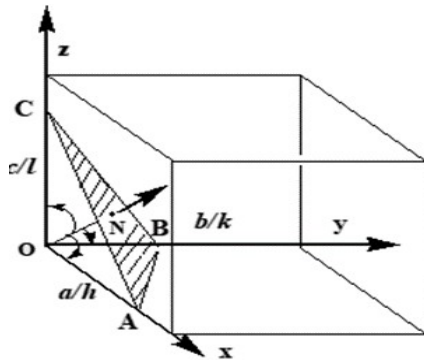


Fig. 1.32: hkl Plane

$ON = d_{hkl}$, Normal to plane (hkl)

$$\cos \alpha = \frac{d_{hkl}}{l(OA)}, \quad \cos \beta = \frac{d_{hkl}}{l(OB)}, \quad \cos \gamma = \frac{d_{hkl}}{l(OC)}$$

Where,

$l(OA) = \frac{a}{h}$, $l(OB) = \frac{b}{k}$, $l(OC) = \frac{c}{l}$ are the intercepts on X, Y and Z axes

$$\cos \alpha = \frac{hd_{hkl}}{a}, \quad \cos \beta = \frac{kd_{hkl}}{b}, \quad \cos \gamma = \frac{ld_{hkl}}{c}$$

Now by the Pythagoras theorem, if the axes are orthogonal,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\left(\frac{h}{a}\right)^2 d_{hkl}^2 + \left(\frac{k}{b}\right)^2 d_{hkl}^2 + \left(\frac{l}{c}\right)^2 d_{hkl}^2 = 1$$

$$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

For cubic crystal system $a=b=c$, therefore $d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

This is the expression for interplanar distance in any (hkl) set of atomic planes and its applicable only to cubic crystal structure.

The direction indicated by vector ON is a direction (hkl) direction and as ON is a perpendicular to an (hkl) plane, we can conclude that - -

Any (hkl) direction is always perpendicular to (hkl).

Q2. What is X-ray Diffraction.

(M.U. May 2013; Dec 2018, 2013,2017) [3 Marks]

The criteria for any kind of diffraction is that the dimension of the diffracting obstacle should be of the order of the wavelength of the beam to be diffracted. The wavelength of X-rays and the atomic separation in a crystal structure are generally of the same order ($\text{\AA}$). Thus, X-ray diffraction is a very important tool in the study of crystal structure.

Q3. Explain Bragg's law.

(M.U. Dec. 2013, 2015, 2016, 2017, 2018; May 2010, 2012, 2013, 2018, June 2019) [8 Marks]

Prof. W.L. Bragg gave a simple interpretation for the intensity pattern obtained when a beam of monochromatic X-rays was made to pass through a crystal.

A crystal may be considered as a stack of parallel planes of atoms. Each plane in a given set has the distribution of atoms. There are several sets of planes oriented in different directions. Each set of layers is rich in atoms differently & also each set is characterized by different inter-planar spacing. Bragg suggested that when a beam of X-rays falls on the crystal, each atom acts as the scattering center and emits secondary X-ray wavelets in all directions whose envelope gives rise to reflected wave-fronts (by Huygens's theory).

So if in a given orientation of crystal, more number of atomic planes rich with atoms are facing the incident X-rays, then we get intensity maxima at different

angles with respect to incident direction. Thus, a very simple design of intensity spots on photographic plate will indicate a simple atomic arrangement in the crystal.

Thus, the X-ray diffraction at crystals may also be considered as Bragg's scattering or Bragg's reflection from certain planes, called Bragg's plane.

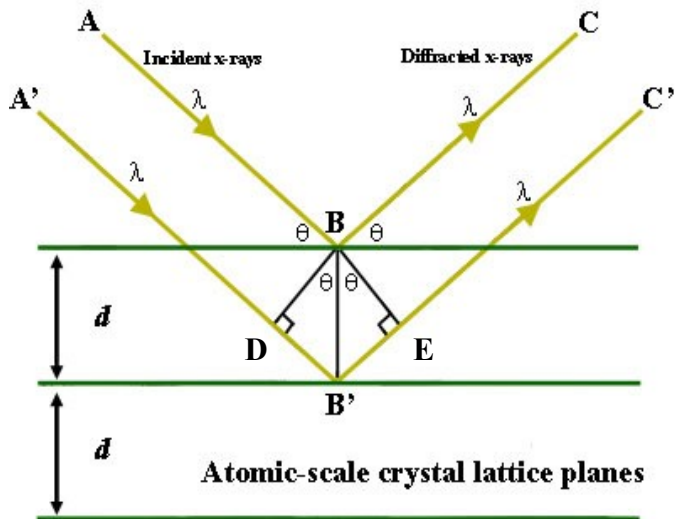


Fig. shows a particular set of atomic planes in a crystal, d is being the interplanar spacing. Suppose an X-ray beam is incident at a glancing angle θ .

Consider two parallel rays are in the incident X-ray beam, AB & $A'B'$ which are reflected by two atoms at points B and B' in adjacent layers. Rays BC & $B'C'$

are reflected rays. Since before points B - D & after points B - E the rays travel the same paths, the path difference between reflected rays, will be $(DB' + B'E)$

From geometry of figure, $\angle DBB' = \angle EBB' = \theta$

Form $\triangle BDB'$ & $\triangle BEB'$, we have $\sin\theta = \frac{B'D}{BB'}$ & $\sin\theta = \frac{B'E}{BB'}$

$$B'D = BB' \sin \theta \quad \wedge \quad B'E = BB' \sin \theta$$

For reflected rays from consecutive planes, path difference = $DB' + B'E = 2 BB' \sin \theta$

But $BB' = d$ (inter-planar spacing), hence path difference = $2d \sin \theta$

For maximum intensity on interference between these reflected rays, we must have

$$2d \sin \theta = n\lambda \quad (n = 1, 2, 3, \dots) \text{ with } \lambda - \text{wavelength of incident X-rays}$$

This relation is known as **Bragg's law**.

The max value of θ is 1. Therefore, Bragg reflection can occur only for wavelength $\lambda < 2d$ and d is of the order of $1\text{\AA}$. This is why we cannot use visible light.

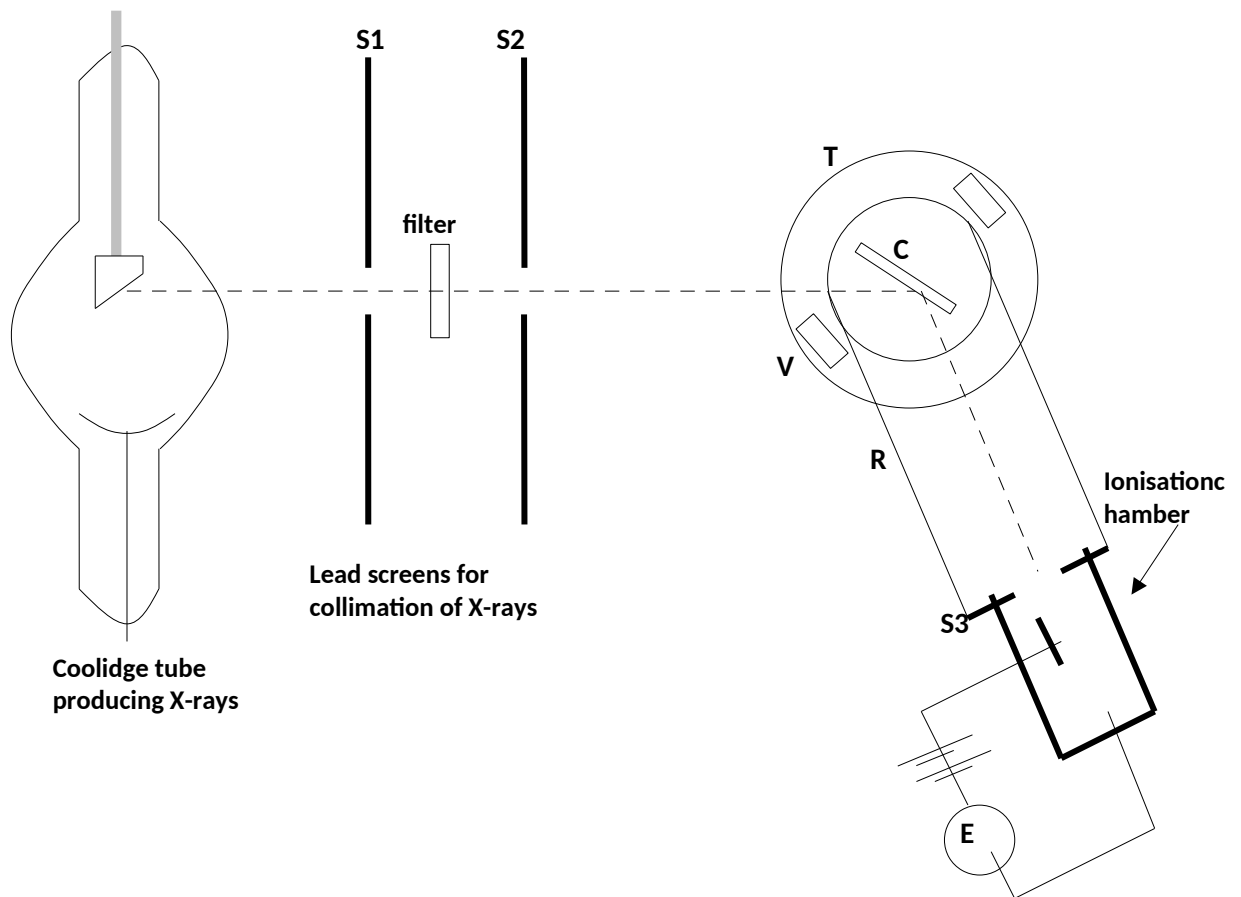
Q4. Explain Bragg's X-ray Spectrometer and its use to analyses crystal structures.

(M.U. Dec. 2013, 2015, 2016, 2017; May 2010, 2012, 2013, 2018; June 2019) [5 Marks]

Bragg devised an apparatus to investigate the structure of single crystal by using X-rays. It is used to measure glancing angle θ . He used crystal as a reflecting grating and not as a transmission grating.

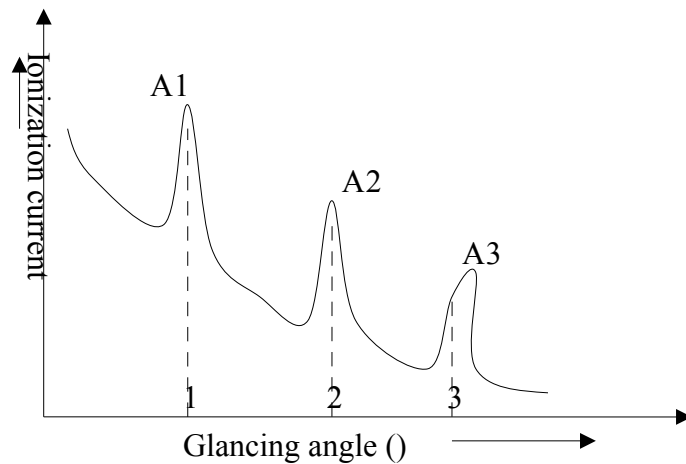
Experiment arrangement is as shown in the above diagram. X-rays from X-ray tube is collimated using two adjustable slits S1 and S2 are made to fall on a crystal C which is mounted in wax on the spectrometer table.

The Vernier scale V is attached to table on which the crystal is placed. The Vernier scale is capable to move on the circular scale S and determine the position of the crystal.



A strong monochromatic X-ray beam is incident on the crystal face. The reflected beam after passing through the slit S_3 enters the ionization chamber mounted on an arm which can be rotated about the same axis as the crystal. Its position can be read by second Vernier V_2 . The gas in the ionization chamber is ionized by the X-ray. The resulting ionization current is measured by the electrometer E, is a measure of intensity of x-rays reflected by the crystal.

The crystal is rotated through small angles (while the arm carrying the ionization current is rotated through double the angles) and the ionization current is measured each time.



The curve of intensity which is the measure of ionization current or ionization current against glancing angle θ is plotted. For a certain value of θ , the intensity of ionization current increases abruptly.

According to Bragg's equation $2d\sin\theta = n\lambda$

For 1st, 2nd and 3rd order values of n are 1, 2, 3 respectively and values of θ are θ_1 , θ_2 , θ_3 respectively.

$$\sin \theta_1 : \sin \theta_2 : \sin \theta_3 = 1 : 2 : 3$$

From the graph, the glancing angles θ_1 , θ_2 , θ_3 can be measured and it can be seen that

$$\sin \theta_1 : \sin \theta_2 : \sin \theta_3 = 1 : 2 : 3$$

From observed values of θ and known values of d & n , the wavelength of the X-ray can be calculated.

Determination of Crystal structure: -

Suppose for the particular crystal used for Bragg's spectrometer, strong reflections from the set of planes (100), (110) and (111) are obtained for angles θ_1 , θ_2 , θ_3 respectively in the first order reflection then the Bragg's equation we have

$$2d_{100}\sin\theta_1 =$$

$$2d_{110}\sin\theta_2 =$$

$$2d_{111} \sin \theta_3 =$$

$$\text{Hence, } d_{100} : d_{110} : d_{111} :: \frac{1}{\sin \theta_1} : \frac{1}{\sin \theta_2} : \frac{1}{\sin \theta_3}$$

Where $\theta_1, \theta_2, \theta_3$ can be obtained from graph Intensity vs angle i.e. where the peak occurs. The reason for selection of planes (100), (110) and (111) is that these are the planes rich enough in terms of atoms.

The ratio of d_{100}, d_{110} & d_{111} for SC, BCC and FCC as follows,

$$\text{SC- } 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$$

$$\text{BCC- } 1 : \frac{1}{\sqrt{2}} : \frac{1}{2\sqrt{3}}$$

$$\text{FCC- } 1 : \frac{1}{2\sqrt{2}} : \frac{1}{\sqrt{3}}$$

Experimentally observed values of $\theta_1, \theta_2, \theta_3$ will provide us d_{100}, d_{110} & d_{111} . By comparing their ratio with above equation, one can determine the crystal structure.

Q5. Define Miller Indices.

(M.U. Dec. 2009, 2013) [5 Marks]

Miller Indices of a particular atomic plane are reciprocals of intercepts of that plane on co-ordinate axis (The intercepts are measured in terms dimensions of the unit cell).